Contents:

Problem 1

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Striker	Forward	Attacking Midfielder	Winger	Total
Players Injured	**45**	**56**	**24**	**20**	**145**
Players Not Injured	**32**	**38**	**11**	**9**	**90**
Total	**77**	**94**	**35**	**29**	**235**

1.1] The probability that a randomly chosen player would suffer an injury

P (Injury) = 145/235 = 0.617 (61.7%)

1.2] The probability that a player is a forward or a winger

P (Forward OR Winger) = (94+29)/235 = 123/235 = 0.523 (52.3%)

1.3] The probability that a randomly chosen player plays in a striker position and has a foot injury

P (Striker AND Injured) = 45/235 = 0.191 (19.1%)

1.4] The probability that a randomly chosen injured player is a striker

This is Conditional Probability.

P (Striker | Injured) = Striker Injured/ Total Injured
= 45/145 = 0.310 (31.0%)

Problem 2

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimetre and a standard deviation of 1.5 kg per sq. centimetre. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain; Answer the questions below based on the given information

2.1] What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?

Since the breaking strength follows a Normal Distribution with mean $\mu=5$ and standard deviation $\sigma=1.5$, all probabilities were calculated using the Normal Cumulative Distribution Function (CDF) where F(x) gives the area under the normal curve up to value x.

$P(X < 3.17) = F(3.17)$

This was obtained directly from the normal CDF, which gives the area to the left of 3.17 under the curve.

Proportion of bags with breaking strength less than 3.17 is 0.1112 (11.12%)

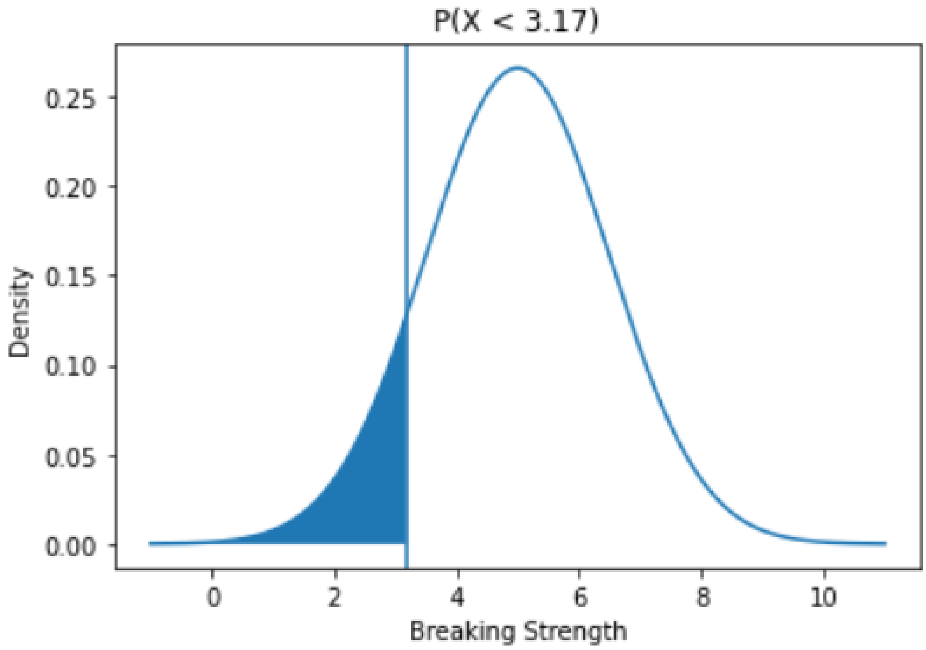

Figure2.1 – Probability Density Function of Breaking Strength representing P(X<3.17)

2.2] What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?

P (X ≥ 3.6) = 1 – F (3.6)

Since the CDF gives area to the left, the right-side probability was found by subtracting from 1.

Proportion of bags with breaking strength of at least 3.6 is 0.8247 (82.47%)

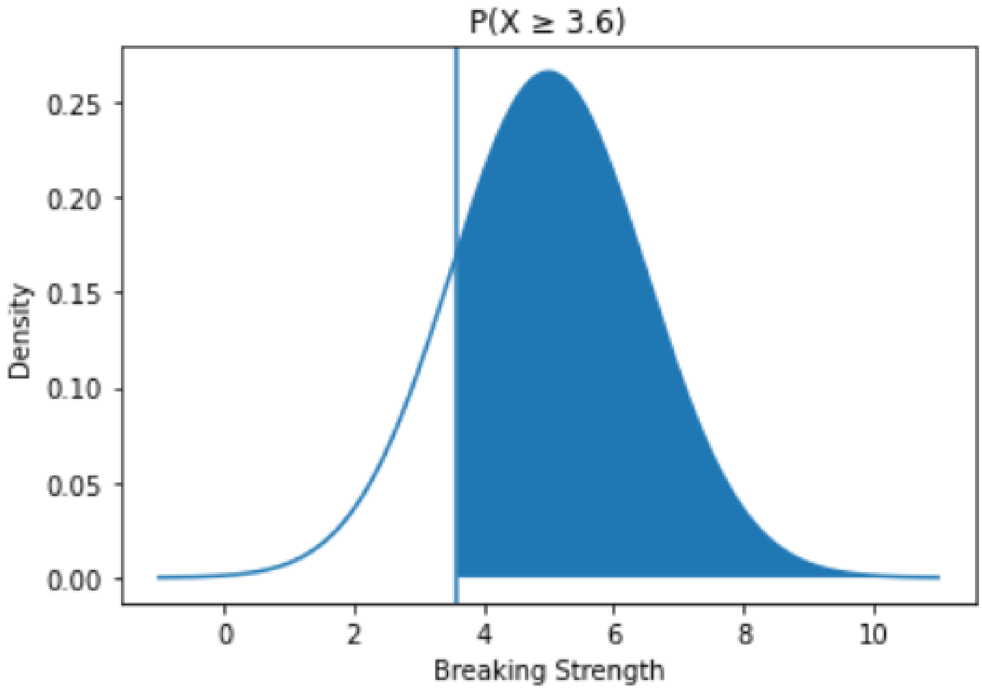

Figure2.2 – Probability Density Function of Breaking Strength representing P(X≥3.6)

2.3] What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?

P (5 ≤ X ≤ 5.5) = F (5.5) – F (5)

The probability between two values was calculated by subtracting the cumulative area at 5 from that at 5.5.

Proportion of bags with breaking strength between 5 and 5.5 is 0.1306 (13.06%)

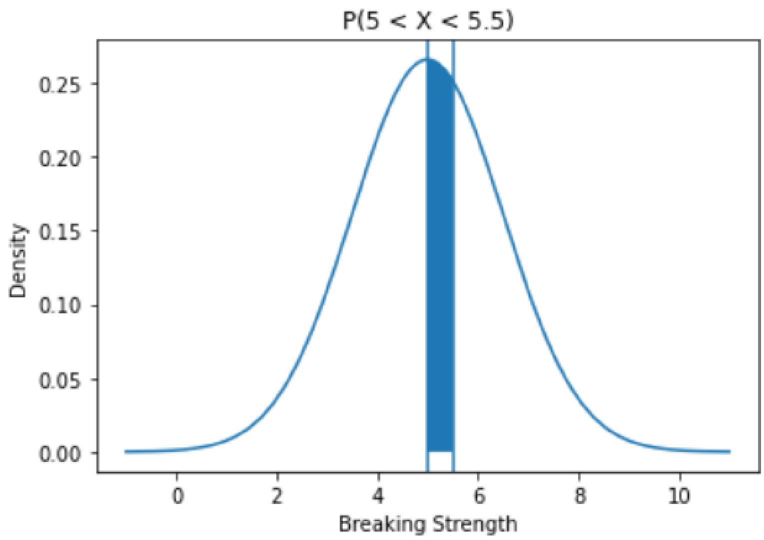

Figure2.3 – Probability Density Function of Breaking Strength representing P (5 ≤ X ≤ 5.5)

2.4] What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?

P (X < 3 or X > 7.5) = F (3) + [1− F (7.5)] = F (3) + [1 – F (7.5)] = F (3) + [1 −F (7.5)]

The total probability outside the range was obtained by adding the left tail area and the right tail area.

Proportion of bags with breaking strength not between 3 and 7.5 is 0.139 (13.9%)

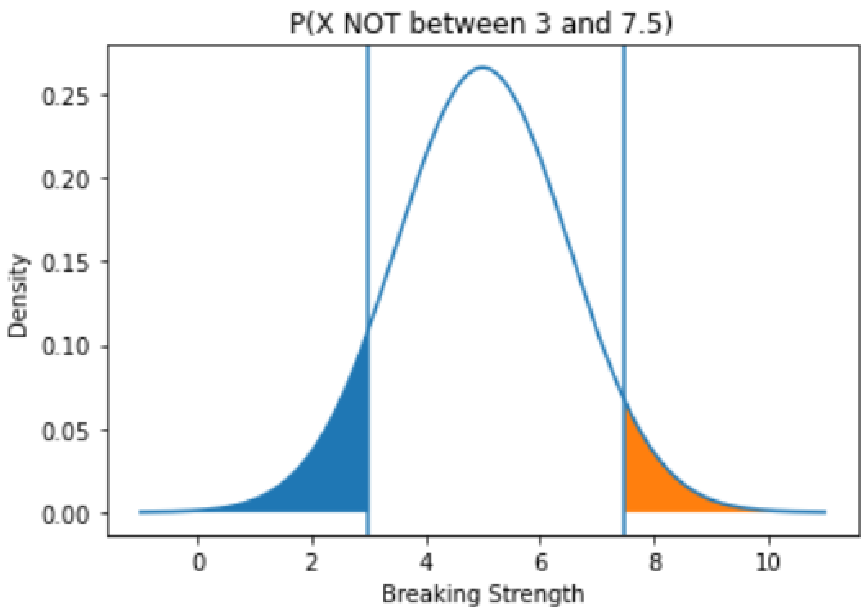

Figure2.4 – Probability Density Function of Breaking Strength representing P (X < 3 or X > 7.5)

Problem 3

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level)

3.1] Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

Test Used: One-Sample t-Test

One-sample t-test was used because the mean hardness of one sample (unpolished stones) is being compared against a known benchmark value (150).

Hypotheses:

- Null Hypothesis (H_0):
  The mean hardness of unpolished stones is greater than or equal to 150.
  H0 : $\mu \geq 150$
- Alternative Hypothesis (H_1):
  The mean hardness of unpolished stones is less than 150.
  Ha : $\mu < 150$
- p-value: 0.0001

Since the p-value was less than 0.05, H_0 was rejected and it was concluded that the mean hardness of unpolished stones is practically lower than 150 and they are not suitable for printing.

3.2] Is the mean hardness of the polished and unpolished stones the same?

Test Used: Two-Sample t-Test

Independent two-sample t-test was used because the means of two independent groups were being compared.

Hypotheses:

- Null Hypothesis (H_0):
  The mean hardness of unpolished stones and polished stones are equal.

H0 : μ1 = μ2

- Alternative Hypothesis (H_1):
  The mean hardness of unpolished stones and polished stones are not equal.
  Ha : μ1 ≠ μ2
- p-value: 0.0016

Since the p-value was less than 0.05, H_0 was rejected and it was concluded that the mean hardness differs significantly between the 2 groups.

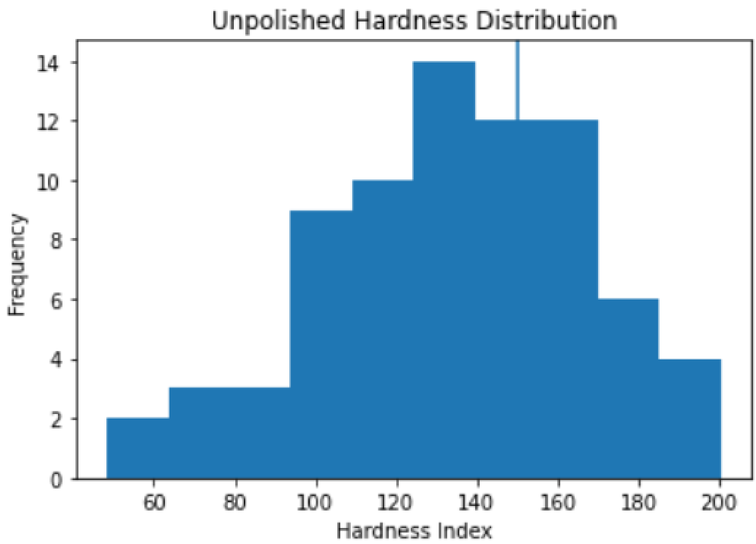

Figure 3.2(a) - Histogram of Hardness Index for Unpolished Stones

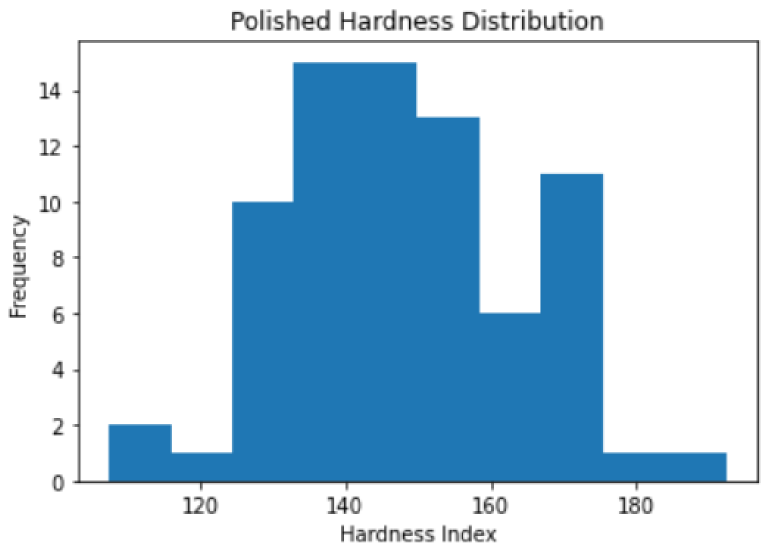

Figure 3.2 (b)- Histogram of Hardness Index for Polished Stones

Problem 4

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favour one method above another and may work better in his/her favourite method. The response is the variable of interest.

4.1] How does the hardness of implants vary depending on dentists?

To determine whether implant hardness varies across dentists, a one-way ANOVA was conducted separately for Alloy 1 and Alloy 2 at a 5% significance level.

Alloy 1 Analysis:

Hypotheses:

- **Null Hypothesis (H_0):**
  The mean implant hardness is the same across all dentists for Alloy 1.
- **Alternative Hypothesis (H_1):**
  At least one dentist has a different mean implant hardness for Alloy 1.

Assumption Checks:

Normality (Shapiro–Wilk Test)

- p-value = 1.19×10^{-5}
- Since $p < 0.05$, we reject the null hypothesis of normality.
- The hardness data for Alloy 1 is not normally distributed.

However, as instructed, we have proceeded with ANOVA despite the violation

Homogeneity of Variance (Levene's Test)

- p-value = 0.2566
- Since $p > 0.05$, we fail to reject the null hypothesis.
- Equal variance assumption holds.

One-Way ANOVA Test:

- ANOVA p-value = **0.1166**

  Since **0.1166 > 0.05**, we fail to reject the null hypothesis.

Conclusion (Alloy 1):
There is no statistically significant evidence that implant hardness differs among dentists when Alloy 1 is used.

Alloy 2 Analysis:

Hypotheses:

- **Null Hypothesis (H_0):**
  The mean implant hardness is the same across all dentists for Alloy 2

- **Alternative Hypothesis (H_1):**
  At least one dentist has a different mean implant hardness for Alloy 2.

Assumption Checks:

Normality (Shapiro–Wilk Test)

- p-value = 0.000403
- Since $p < 0.05$, the normality assumption is violated.

As required, we proceed with ANOVA regardless.

Homogeneity of Variance (Levene's Test)

- p-value = 0.2369
- Since $p > 0.05$, equal variance assumption holds.

One-Way ANOVA Test:

- ANOVA p-value = 0.7180
- Since $0.7180 > 0.05$, we fail to reject the null hypothesis.

Conclusion (Alloy 2):

- There is no statistically significant evidence that implant hardness differs among dentists when Alloy 2 is used.

Overall Business Interpretation:

- For both Alloy 1 and Alloy 2, implant hardness does not significantly vary across dentists.
- This suggests that implant hardness is not dependent on the dentist performing the procedure for either alloy.

- From an operational perspective, dentist variability does not appear to influence implant hardness outcomes.

4.2] How does the hardness of implants vary depending on methods?

To determine whether implant hardness varies across different implant methods, a one-way ANOVA was conducted separately for Alloy 1 and Alloy 2 at a 5% significance level.

Alloy 1 Analysis:

Hypotheses:

- **Null Hypothesis (H_0):**
  The mean implant hardness is the same across all methods for Alloy 1.
- **Alternative Hypothesis (H_1):**
  At least one method has a different mean implant hardness for Alloy 1.

Assumption Checks:

Normality (Shapiro–Wilk Test)

- p-value = 1.19×10^{-5}
- Since $p < 0.05$, the normality assumption is violated.

Homogeneity of Variance (Levene's Test)

- p-value = 0.00342
- Since $p < 0.05$, the equal variance assumption is also violated.

As instructed, the ANOVA test was still conducted despite assumption violations.

One-Way ANOVA Test:

- ANOVA p-value = 0.00416
- Since $0.00416 < 0.05$, we reject the null hypothesis.

Conclusion (Alloy 1):

- There is statistically significant evidence that implant hardness differs among methods when Alloy 1 is used.
- This indicates that the choice of method plays a significant role in determining hardness outcomes for Alloy 1.

Alloy 2 Analysis:

Hypotheses:

- **Null Hypothesis (H_0):**
  The mean implant hardness is the same across all methods for Alloy 2.
- **Alternative Hypothesis (H_1):**
  At least one method has a different mean implant hardness for Alloy 2.

Assumption Checks:

Normality (Shapiro–Wilk Test)

- p-value = 0.000403
- Since $p < 0.05$, the normality assumption is violated.

Homogeneity of Variance (Levene's Test)

- p-value = 0.04469
- Since $p < 0.05$, the equal variance assumption is also violated.

As instructed, the ANOVA test was still conducted despite assumption violations.

One-Way ANOVA Test:

- ANOVA p-value = 5.42×10^{-6}
- Since $0.00416 < 0.05$, we reject the null hypothesis.

Conclusion (Alloy 2):

- There is strong statistical evidence that implant hardness differs among methods when Alloy 2 is used.
- The very small p-value indicates a highly significant method effect.

Overall Business Interpretation:

- For both Alloy 1 and Alloy 2, implant hardness varies significantly depending on the method used.
- The effect is particularly strong for Alloy 2.
- Method selection is therefore a critical factor in determining implant hardness performance.
- Standardizing or optimizing method selection could improve consistency and quality outcomes.

4.3] What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

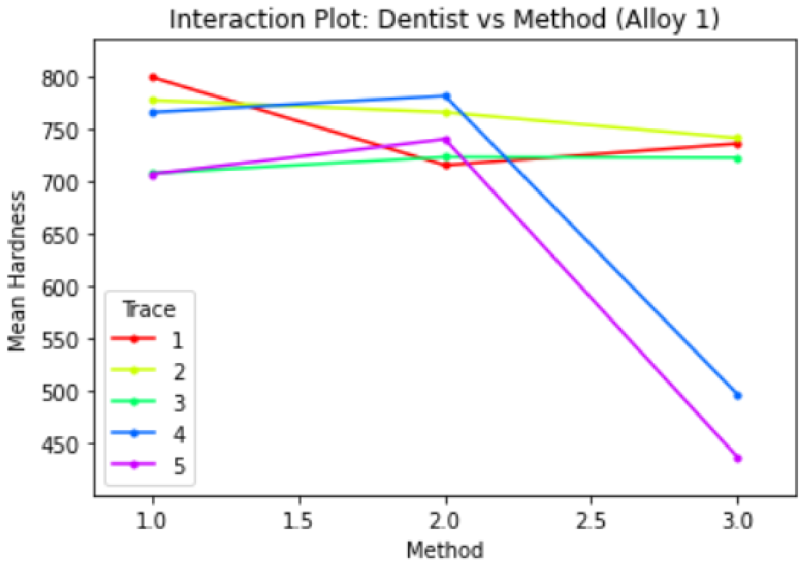

Figure 4.3 (a) Dentist–Method Interaction Plot for Alloy1

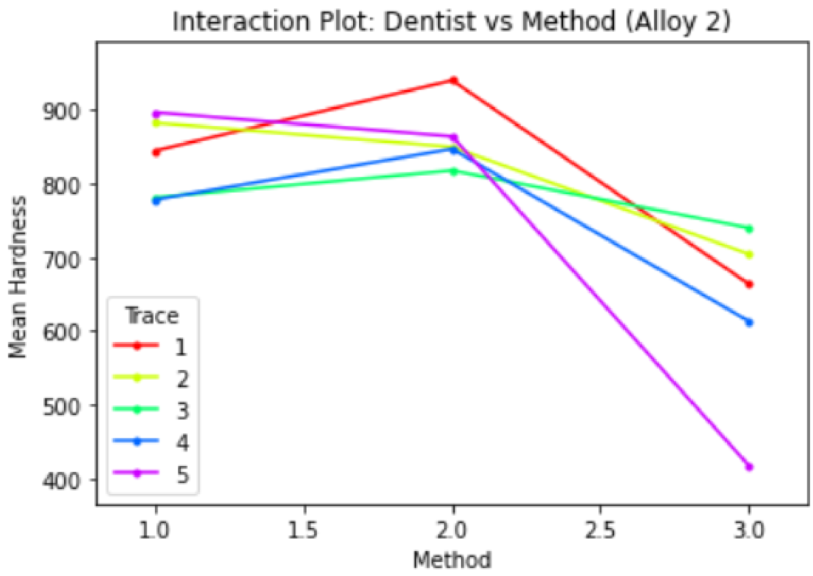

Figure 4.3 (b) Dentist–Method Interaction Plot for Alloy2

Overall Business Interpretation:

The interaction plots indicate that implant hardness is strongly influenced by the choice of method, with Method 3 showing substantially lower hardness values, particularly for certain dentists. The non-parallel trends suggest that method effectiveness varies by dentist. Standardizing toward more consistent-performing methods (such as Method 2) may improve overall implant quality and reduce variability.

4.4] How does the hardness of implants vary depending on dentists and methods together?

A two-way ANOVA with interaction was conducted separately for Alloy 1 and Alloy 2 at a 5% significance level to examine:

- The main effect of Dentist
- The main effect of Method
- The interaction effect between Dentist and Method

Alloy 1 Analysis:

Hypotheses:

- **Null Hypothesis (H_0):**
  H_0 (Dentist): Mean hardness is equal across dentists.
  H_0 (Method): Mean hardness is equal across methods.
  H_0 (Interaction): There is no interaction between dentist and method.
- **Alternative Hypothesis (H_1):**
  H_1 (Interaction): The effect of method depends on the dentist.

Assumption Checks:

- Shapiro (Normality) p-value = 0.0902, Assumption satisfied
- Levene (Equal Variance) p-value = 0.3128, Assumption satisfied

Two-Way ANOVA Results (Alloy 1):

Effect	p-value	Interpretation
Dentist	0.011484	Significant
Method	0.000284	Significant
Interaction	0.006793	Significant

Since all p-values are less than 0.05:

- Implant hardness varies significantly across dentists.
- Implant hardness varies significantly across methods.
- There is a significant interaction between dentist and method.

Business Interpretation: (Alloy1)

The significant interaction ($p = 0.006793$) indicates that the effect of method on hardness depends on which dentist performs the procedure. The interaction plot suggests that most significant differences involve:

- Dentist 4 – Method 3
- Dentist 5 – Method 3

These combinations produced significantly lower hardness values compared to many other dentist–method combinations.

Alloy 2 Analysis:

Hypotheses:

- **Null Hypothesis (H_0):**
  H_0 (Dentist): Mean hardness is equal across dentists.
  H_0 (Method): Mean hardness is equal across methods.
  H_0 (Interaction): There is no interaction between dentist and method.
- **Alternative Hypothesis (H_1):**
  H_1 (Interaction): The effect of method depends on the dentist.

Assumption Checks:

- Shapiro (Normality) p-value = 0.0946, Assumption satisfied
- Levene (Equal Variance) p-value = 0.7832, Assumption satisfied

Two-Way ANOVA Results (Alloy 2):

Effect	**p-value**	**Interpretation**
Dentist	**0.371833**	Not Significant
Method	**0.000004**	Significant
Interaction	**0.093234**	Not Significant

- Hardness significantly varies across methods.
- Hardness does not significantly vary across dentists.
- There is no statistically significant interaction at the 5% level.

Overall Business Interpretation:

- Method selection is the strongest driver of implant hardness for both alloys.
- For Alloy 1, hardness depends on both dentist and method, and their interaction.
- For Alloy 2, hardness depends primarily on method, not dentist.
- Method 3 appears to produce substantially lower hardness in several cases, particularly for Dentist 4 and 5.